

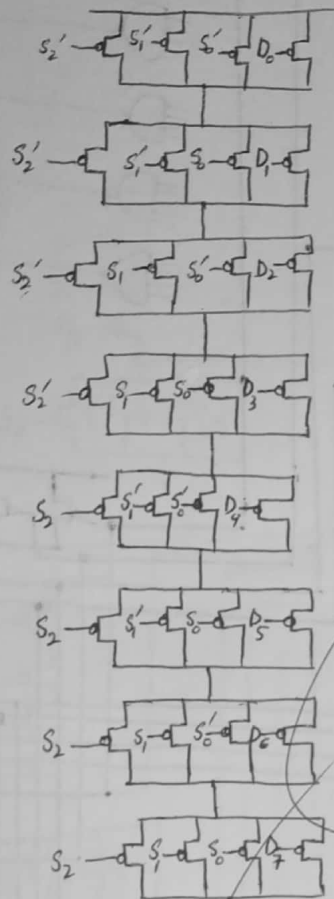
Ayka Asim

FA17-BCE-021-A

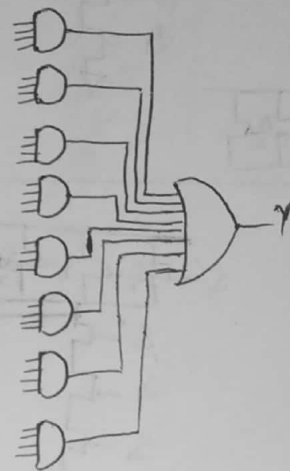
Assignment # 1 - Very large scale integration design

Question # 1: Design a set of CMOS circuits for an 8X1 Mux with the following techniques:

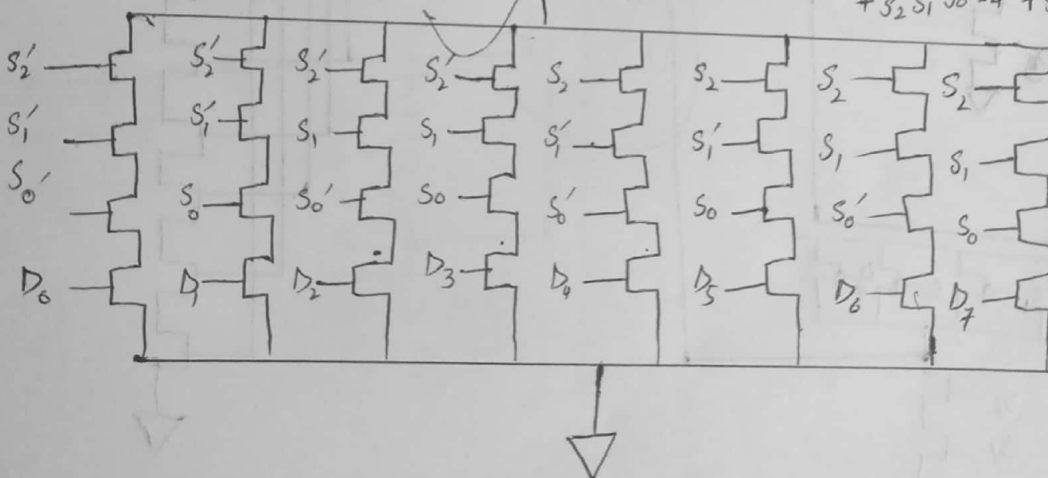
a) Compound CMOS logic implementation



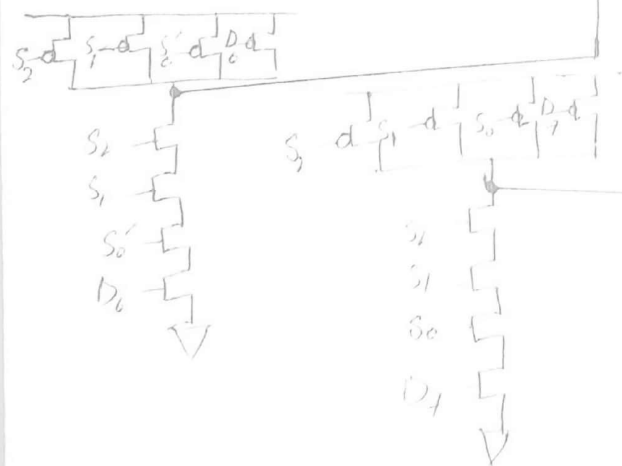
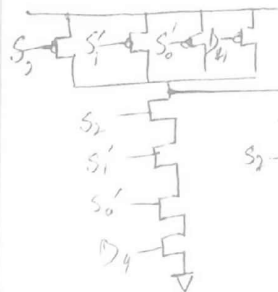
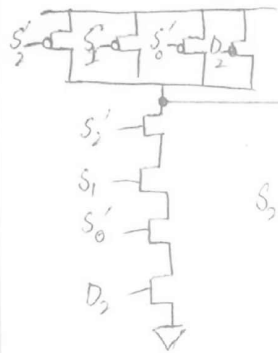
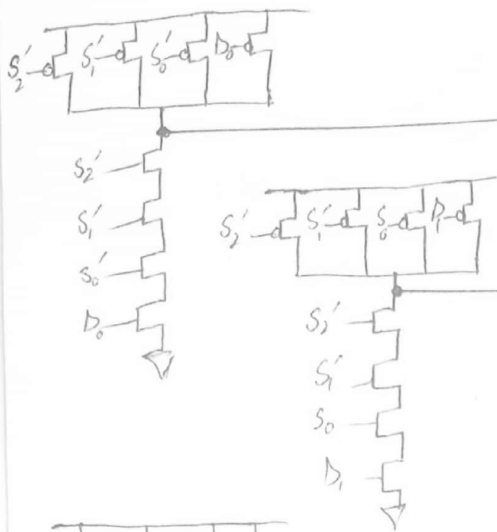
• Gates implementation



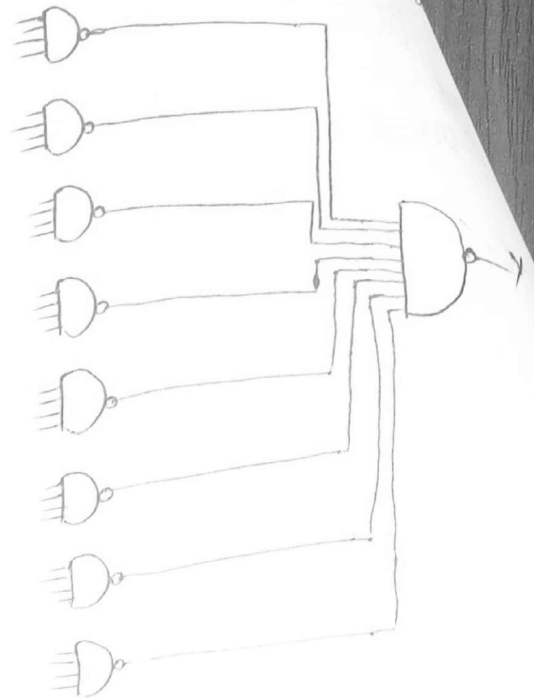
$$Y = S_2' S_1' S_0' D_0 + S_2' S_1' S_0 D_1 + S_2' S_1 S_0' D_2 + S_2' S_1 S_0 D_3 + S_2 S_1' S_0' D_4 + S_2 S_1' S_0 D_5 + S_2 S_1 S_0' D_6 + S_2 S_1 S_0 D_7$$



b) CMOS NAND gates

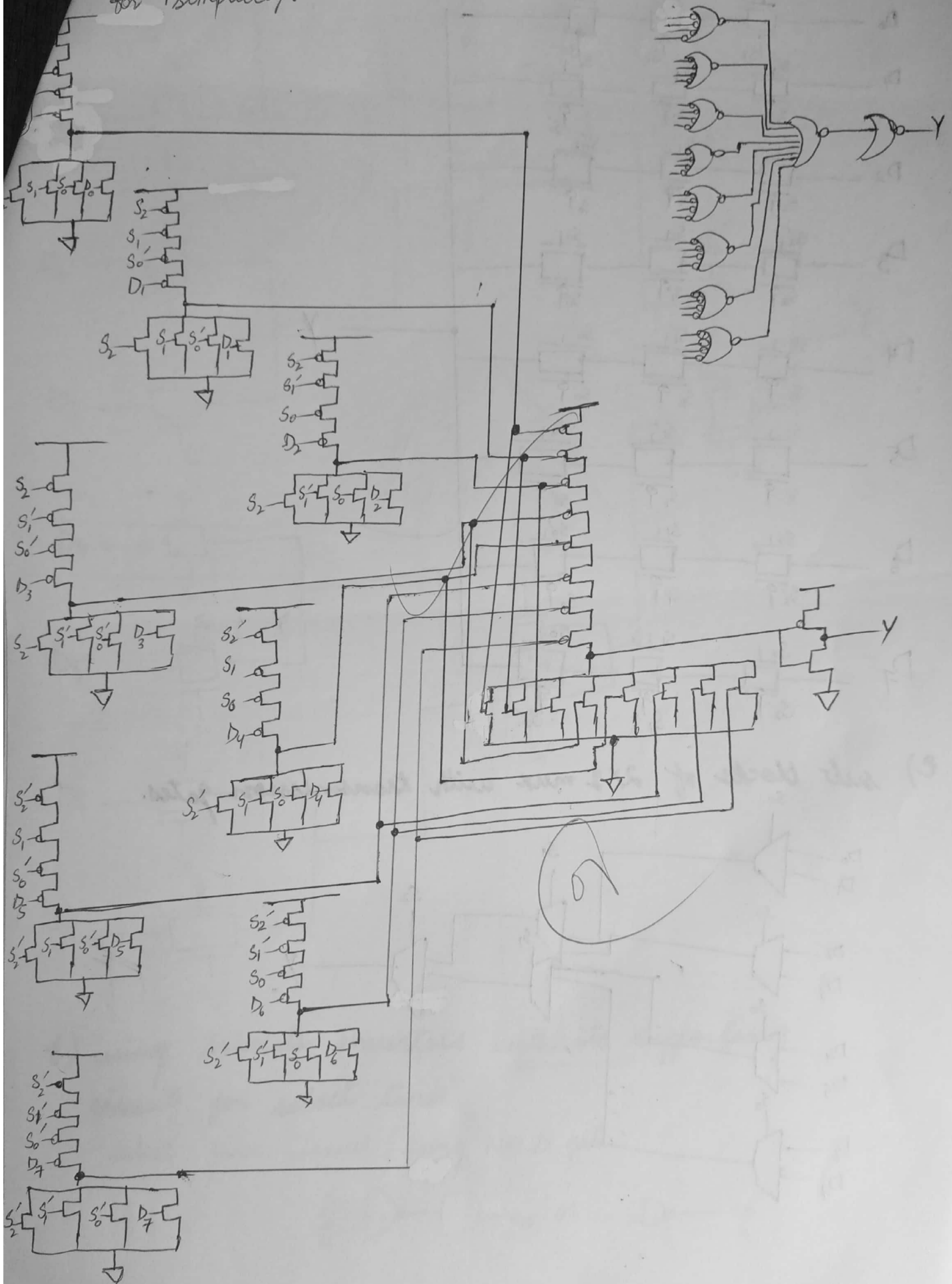


• NAND gates implementation



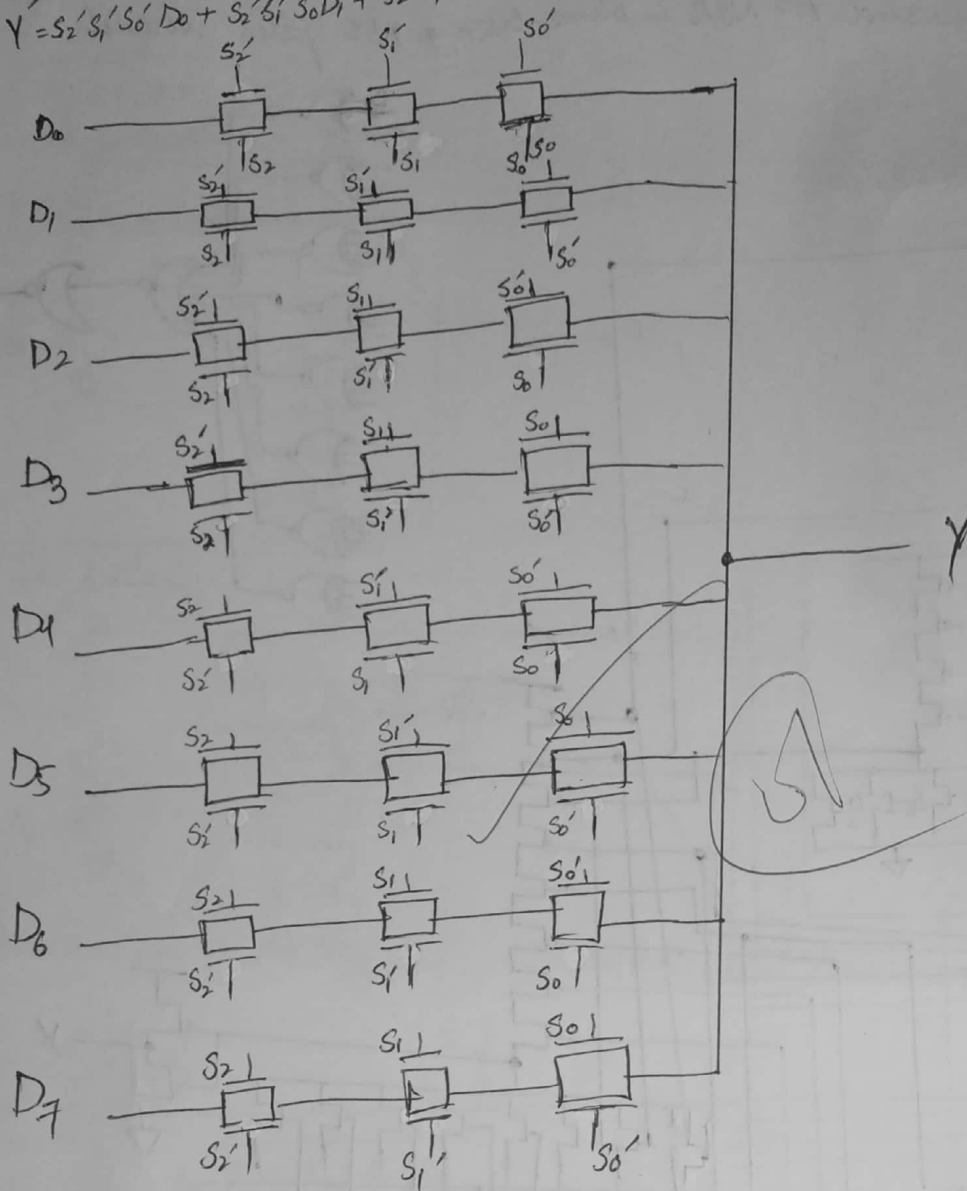
nor gates implementation

The inputs are inverted for NOR implementation • nor gates circuit for simplicity.

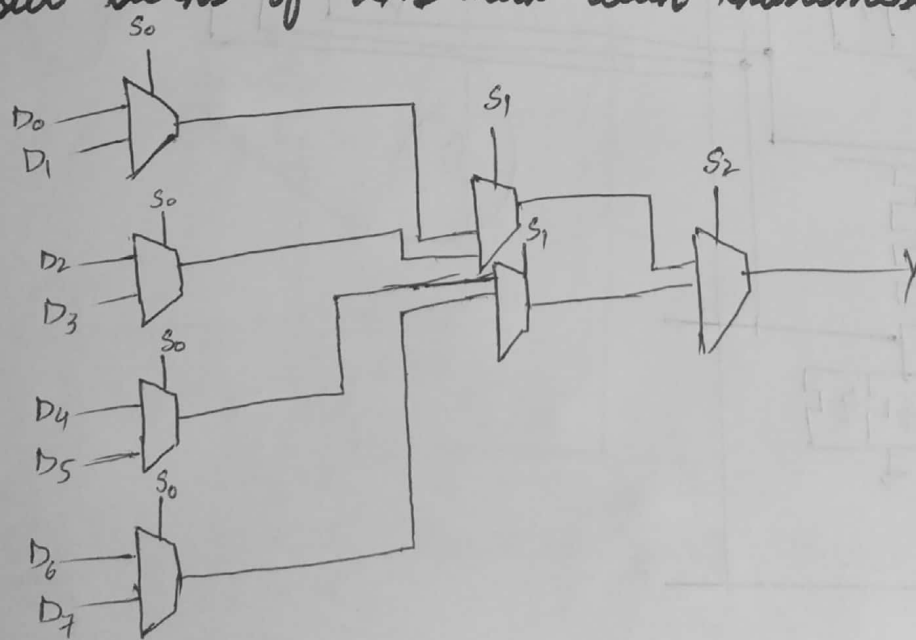


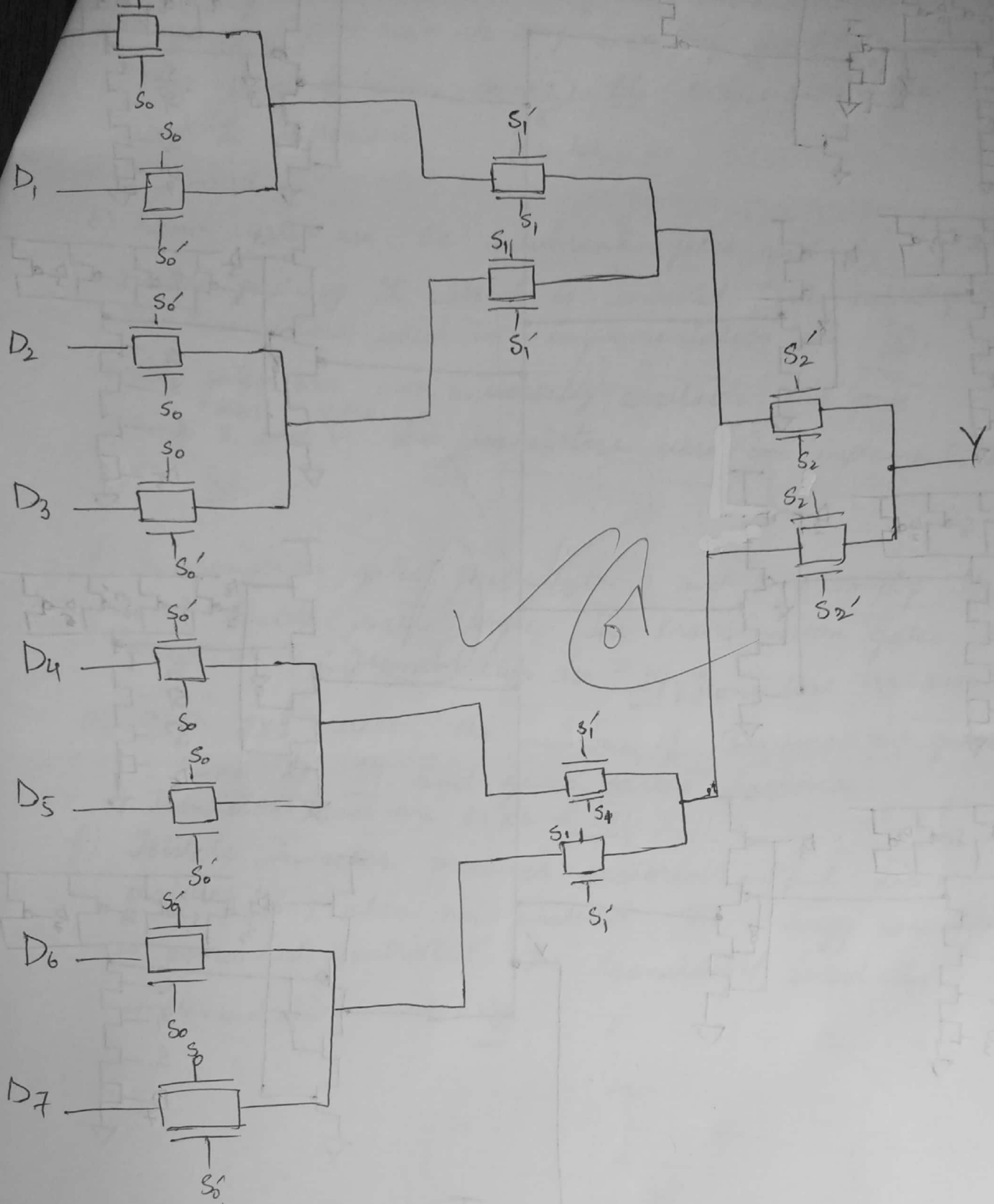
d) using only transmission gates -

$$Y = s_2' s_1' s_0' D_0 + s_2' s_1' s_0 D_1 + s_2' s_1 s_0' D_2 + s_2' s_1 s_0 D_3 + s_2 s_1' s_0' D_4 + s_2 s_1' s_0 D_5 + s_2 s_1 s_0' D_6 + s_2 s_1 D_7$$



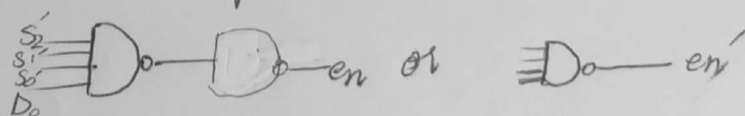
e) sub blocks of 2x1 mux with transmission gates.

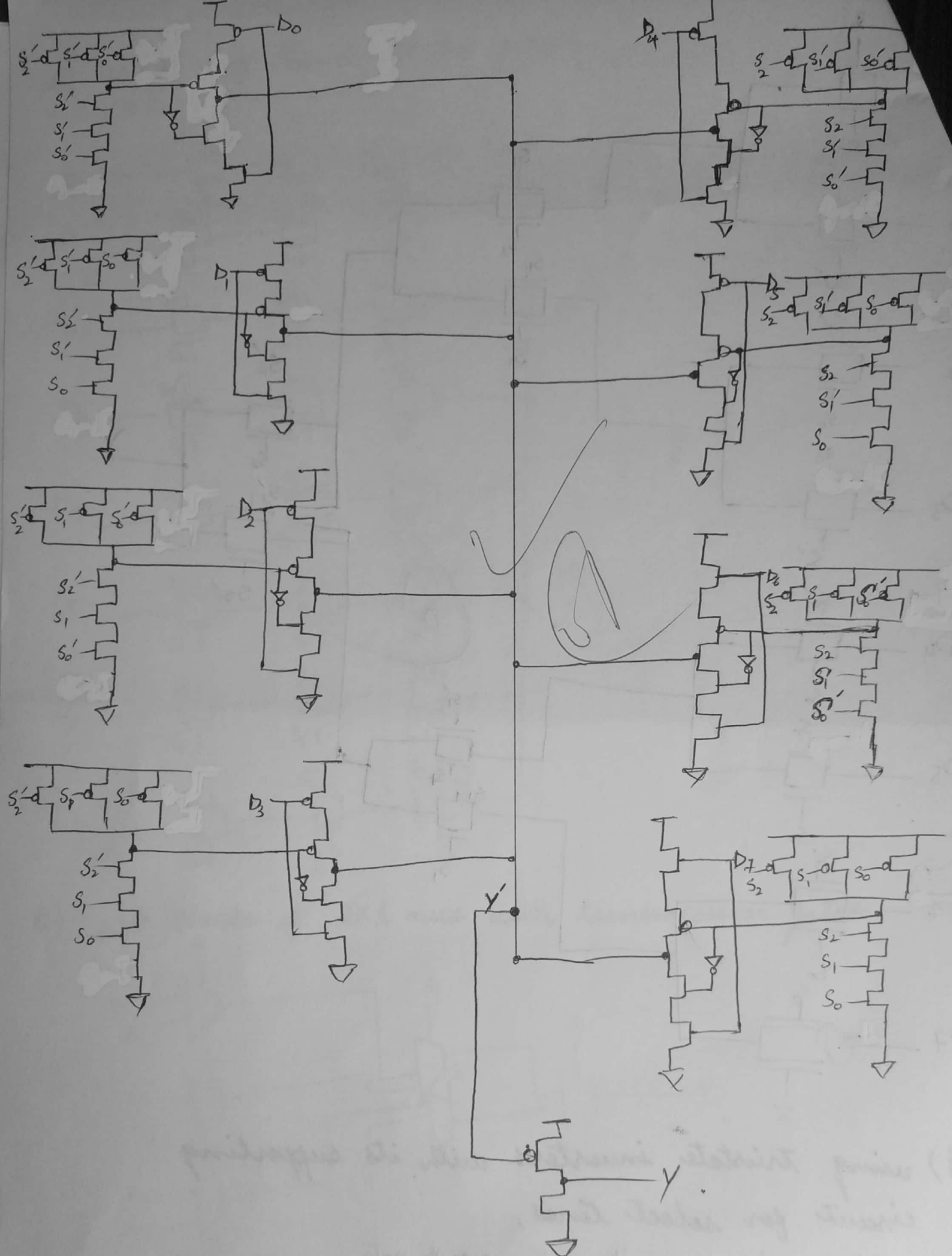




8) using tri-state inverters with its supporting circuit for select lines.

select line circuit using NAND gate.





n #2 - The pros and cons of each implementation

a) compound gates can do any inverting function. In above implementation $64 + (2) = 66$ transistors are used to implement inverter
8X1 mux.

b) Nand gates are the universal gates and by bubble pushing it circuit is converted into nand gates. The #transistors used in implementation are 80.

c) NOR gates are also universally available and pass strong ^{PMOS} 1 and ^{NMOS} 0. The transistors used in implementation are 82.

d) Transmission gates pass both 0 and 1 strongly. It uses double rail logic. The transmission gates used in implementation are 24 Transistors are $24 \times 2 = 48$

e) with 2X1 muxes, the number of transmission gates reduces to 14 and gives faster response. Transistors used are $14 \times 2 = 28$

f) Tristate inverter produces restored output. also produces (Z) when not enabled. The energy consumption is somewhat controlled. The transistors used in implementation are 96.